Chapter 1

EXERCISE 1 **Determine whether each sequence is convergent or divergent.** If it is convergent, give its limit.

1. $1, 3, 5, 7, \ldots$
2. $3.49, 3.499, 3.4999, 3.49999, \ldots$
3. $\frac{1}{10}, -\frac{1}{100}, \frac{1}{1000}, -\frac{1}{10\,000}, \ldots$
4. $\frac{20}{27}, \frac{121}{162}, \frac{182}{243}, \frac{1093}{1458}, \frac{1640}{2187}, \ldots$
5. $3, 4, 3, 4, 3, 4, \ldots$

EXERCISE 2

1. Evaluate :
   a. $\lim\limits_{x\to 3}(x+4)$
   b. $\lim\limits_{x\to -1}(5-2x)$
   c. $\lim\limits_{x\to 4}(3x-1)$
   d. $\lim\limits_{x\to 2}(5x^2-3x+2)$
   e. $\lim\limits_{h\to 0} h^2(1-h)$
   f. $\lim\limits_{x\to 0}(x^2+5)$
2. Evaluate :
   a. $\lim\limits_{x\to 0} 5$
   b. $\lim\limits_{h\to 2} 7$
   c. $\lim\limits_{x\to 0} c$ (where c is a constant)
3. Evaluate :
   a. $\lim\limits_{x\to 1}\dfrac{x^2-3x}{x}$
   b. $\lim\limits_{h\to 2}\dfrac{h^2+5h}{h}$
   c. $\lim\limits_{x\to 0}\dfrac{x-1}{x+1}$
   d. $\lim\limits_{x\to 0}\dfrac{x}{x}$
4. Evaluate the following limits :
   a. $\lim\limits_{x\to 0}\dfrac{x^2-3x}{x}$
   b. $\lim\limits_{x\to 0}\dfrac{x^2+5x}{x}$
   c. $\lim\limits_{x\to 0}\dfrac{2x^2-x}{x}$
   d. $\lim\limits_{h\to 0}\dfrac{2h^2+6h}{h}$
   e. $\lim\limits_{h\to 0}\dfrac{3h^2-4h}{h}$
   f. $\lim\limits_{h\to 0}\dfrac{h^3-8h}{h}$
   g. $\lim\limits_{x\to 1}\dfrac{x^2-x}{x-1}$
   h. $\lim\limits_{x\to 2}\dfrac{x^2-2x}{x-2}$
   i. $\lim\limits_{x\to 3}\dfrac{x^2-x-6}{x-3}$

EXERCISE 3

1. For each of the following functions :
   i discuss the behaviour near the asymptotes and hence deduce their equations ;
   ii state the values of $\lim\limits_{x\to-\infty} f(x)$ and $\lim\limits_{x\to\infty} f(x)$.

a. $f(x) = \dfrac{1}{x}$

b. $f(x) = \dfrac{3x-2}{x+3}$

c. $f(x) = \dfrac{1-2x}{3x+2}$

d. $f(x) = \dfrac{x}{1-x}$

e. $f(x) = \dfrac{x^2-1}{x^2+1}$

f. $f(x) = \dfrac{x}{x^2+1}$

2. **a.** Sketch the graph of $y = e^x - 6$.

b. Hence discuss the value and geometric interpretation of :

i. $\lim\limits_{x\to-\infty}(e^x-6)$

ii. $\lim\limits_{x\to\infty}(e^x-6)$

3. Find, if possible, $\lim\limits_{x\to-\infty}(2e^{-x}-3)$ and $\lim\limits_{x\to\infty}(2e^{-x}-3)$.

EXERCISE 4

1. Write an expression for the gradient of a secant line for each function and simplify :

a. $f(x) = 3x + 4$

b. $f(x) = 2x^2 - 1$

c. $f(x) = x^2 + 2x + 3$

EXERCISE 5 **Use the first principles formula** to find the gradient of the tangent to :

1. $f(x) = 3x + 5$ at $x = -2$

2. $f(x) = 5 - 2x^2$ at $x = 3$

3. $f(x) = x^2 + 3x - 4$ at $x = 3$

4. $f(x) = 5 - 2x - 3x^2$ at $x = -2$

EXERCISE 6

1. **Differentiate from first principles :**

a. x^2

b. x^3

c. x^4

2. Consider the binomial expansion :

$$(x+h)^n = \binom{n}{0}x^n + \binom{n}{1}x^{n-1}h + \binom{n}{2}x^{n-2}h^2 + \cdots + \binom{n}{n}h^n.$$

Use $f'(x) = \lim\limits_{h\to 0}\dfrac{f(x+h)-f(x)}{h}$ to find the derivative of $f(x) = x^n$ for $n \in \mathbb{N}$.

3. From first principles find the derivatives of $4x^2$ and $2x^3$.

By comparison with 1., copy and complete : *"If* $f(x) = cx^n$*, then* $f'(x) = \ldots$*"*

EXERCISE 7

1. Find $f'(x)$ given that $f(x)$ is :

a. x^3

b. $2x^3$

c. $7x^2$

d. $6\sqrt{x}$

e. $3\sqrt[3]{x}$

f. $x^2 + x$

g. $4 - 2x^2$

h. $x^2 + 3x - 5$

i. $\dfrac{1}{2}x^4 - 6x^2$

j. $\dfrac{3x-6}{x}$

k. $\dfrac{2x-3}{x^2}$

l. $\dfrac{x^3+5}{x}$

m. $\dfrac{x^3+x-3}{x}$

n. $\dfrac{1}{\sqrt{x}}$

o. $(2x-1)^2$

p. $(x+2)^3$

2. Find $\dfrac{dy}{dx}$ for :

a. $y = 2.5x^3 - 1.4x^2 - 1.3$

b. $y = \pi x^2$

c. $y = \dfrac{1}{5x^2}$

d. $y = 100x$

e. $y = 10(x+1)$

f. $y = 4\pi x^3$

EXERCISE 8

1. Use the product rule to differentiate :

a. $f(x) = x(x-1)$

b. $f(x) = 2x(x+1)$

2. Find $\frac{dy}{dx}$ using the product rule :

a. $y = x^2(2x-1)$

b. $y = 4x(2x+1)^3$

c. $y = \sqrt{x}(x-3)^2$

d. $y = 5x^2(3x^2-1)^2$

EXERCISE 9

1. Use the quotient rule to find $\frac{dy}{dx}$ if :

a. $y = \frac{1+3x}{x^2+1}$

b. $y = \frac{x^2}{2x+1}$

c. $y = \frac{x}{x^2-3}$

d. $y = \frac{\sqrt{x}}{2x}$

e. $y = \frac{x^2-3}{3x-x^2}$

f. $y = \frac{x}{\sqrt{1-3x}}$

2. Find the gradient of the tangent to :

a. $y = \frac{x}{1-2x}$ at $x = 1$

b. $y = \frac{x^3}{2x+1}$ at $x = -1$

c. $y = \frac{\sqrt{x}}{2x+1}$ at $x = 4$

d. $y = \frac{x^2}{\sqrt{x^2+5}}$ at $x = -2$

Check your answers using technology.

(3a) If $y = \frac{2\sqrt{x}}{1-x}$, show that

$$\frac{dy}{dx} = \frac{x+1}{\sqrt{x}(1-x)^2}.$$

(3b) For what values of x is $\frac{dy}{dx}$ i) zero ii) undefined ?

EXERCISE 10

1. Write in the form au^n, clearly stating what u is :
   - **a.** $\frac{1}{(2x-1)^2}$
   - **b.** $\sqrt{x^2-3x}$
   - **c.** $\frac{2}{\sqrt{2-x^2}}$
   - **d.** $\sqrt[3]{x^3-x^2}$
   - **e.** $\frac{4}{(3-x)^3}$
   - **f.** $\frac{10}{x^2-3}$
2. Find the gradient function $\frac{dy}{dx}$ for :
   - **a.** $y=(4x-5)^2$
   - **b.** $y=\frac{1}{5-2x}$
   - **c.** $y=\sqrt{3x-x^2}$
   - **d.** $y=(1-3x)^4$
   - **e.** $y=6(5-x)^3$
   - **f.** $y=\sqrt[3]{2x^3-x^2}$
3. Find the gradient of the tangent to :
   - **a.** $y=\sqrt{1-x^2}$ at $x=\frac{1}{2}$
   - **b.** $y=(3x+2)^6$ at $x=-1$
   - **c.** $y=\frac{1}{(2x-1)^4}$ at $x=1$
   - **d.** $y=6\times\sqrt[3]{1-2x}$ at $x=0$
   - **e.** $y=\frac{4}{x+2\sqrt{x}}$ at $x=4$
   - **f.** $y=\left(x+\frac{1}{x}\right)^3$ at $x=1$

EXERCISE 11

Find the gradient function for $f(x)$ equal to :

a) e^{4x}	b) e^x+3	c) $\exp(-2x)$	d) $e^{\frac{x}{2}}$
e) $2e^{-\frac{x}{2}}$	f) $1-2e^{-x}$	g) $4e^{\frac{x}{2}}-3e^{-x}$	h) $\frac{e^x+e^{-x}}{2}$
i) e^{-x^2}	j) $e^{\frac{1}{x}}$	k) $10(1+e^{2x})$	l) $20(1-e^{-2x})$
m) e^{2x+1}	n) $e^{\frac{\pi}{4}x}$	o) e^{1-2x^2}	p) $e^{-0.02x}$

EXERCISE 12

Find the derivative of :

a) xe^x b) x^3e^{-x} c) $\frac{e^x}{x}$ d) $\frac{x}{e^x}$

e) x^2e^{3x} f) $\frac{e^x}{\sqrt{x}}$ g) $\sqrt{x}e^{-x}$ h) $\frac{e^x+2}{e^{-x}+1}$

EXERCISE 13

Find the gradient of the tangent to :

a) $y = (e^x + 2)^4$ at $x = 0$

b) $y = \frac{1}{2 - e^{-x}}$ at $x = 0$

c) $y = \sqrt{e^{2x} + 10}$ at $x = \ln 3$

Check your answers using technology.

EXERCISE 14

Given $f(x) = e^{kx} + x$ and $f'(0) = -8$, find k.

EXERCISE 15

Find the gradient function of :

a) $y = \ln(7x)$ b) $y = \ln(2x + 1)$ c) $y = \ln(x - x^2)$

d) $y = 3 - 2\ln x$ e) $y = x^2 \ln x$ f) $y = \frac{\ln x}{2x}$

g) $y = e^x \ln x$ h) $y = (\ln x)^2$ i) $y = \sqrt{\ln x}$

j) $y = e^{-x} \ln x$ k) $y = \sqrt{x}\ln(2x)$ l) $y = \frac{2\sqrt{x}}{\ln x}$

m) $y = 3 - 4\ln(1 - x)$ n) $y = x\ln(x^2 + 1)$

EXERCISE 16

Find $\frac{dy}{dx}$ for :

a) $y = x \ln 5$ b) $y = \ln(x^3)$ c) $y = \ln(x^4 + x)$

d) $y = \ln(10 - 5x)$ e) $y = [\ln(2x + 1)]^3$ f) $y = \ln(4x)$

g) $y = \ln\left(\frac{1}{x}\right)$ h) $y = x\ln(\ln x)$ i) $y = \frac{1}{\ln x}$

EXERCISE 17

Find $f''(x)$ given that :

a) $f(x) = 3x^2 - 6x + 2$

b) $f(x) = \dfrac{2}{\sqrt{x}} - 1$

c) $f(x) = 2x^3 - 3x^2 - x + 5$

d) $f(x) = \dfrac{2 - 3x}{x^2}$

e) $f(x) = (1 - 2x)^3$

f) $f(x) = \dfrac{x + 2}{2x - 1}$